ASSIGNMENT

Can the Following Tasks Currently Be Solved by Computers?

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Course Code: CSA1718

Summary

No.	Task	Can computers do it today?	Status
1	Playing a decent game of table tennis	Yes	Achieved (elite amateur level)
2	Discovering and proving a new mathematical theorem	Yes, in part	Achieved in specific areas
3	Writing an intentionally funny story	Partly	Possible, but quality is inconsistent
4	Translating spoken English into spoken French	Yes	Achieved (real-time systems exist)
5	Performing a complex surgical operation	Not fully	Only in lab settings, not on live patients

1. Playing a Decent Game of Table Tennis

Answer: Yes. Computers (robots) can now play table tennis at a level well above "decent".

Table tennis robots have existed since the 1980s, but for decades they could hold rallies only at amateur level. In April 2026, Sony AI published a study in Nature on a robot called Ace. Using cameras that track the ball's position, speed and spin, and a learned control policy, Ace won three of five matches against elite players under official rules. It lost both matches against professional players, though it won one game against a professional.

Conclusion: the task is solved at a good-to-elite amateur level. Beating the very best professionals consistently is still being worked on.

2. Discovering and Proving a New Mathematical Theorem

Answer: Yes, in specific areas. (The question says "game"; this is the standard textbook exercise on discovering and proving a new mathematical theorem.)

- **Automated theorem proving:** In 1996 the program EQP proved the Robbins conjecture, an open problem in algebra that human mathematicians had not solved. The Four Colour Theorem (1976) was also proved with essential computer assistance.
- **Modern AI:** DeepMind's AlphaProof and AlphaGeometry solved International Mathematical Olympiad problems at silver-medal level in 2024, and AI systems reached gold-medal standard in 2025.
- **Discovery:** Systems such as AlphaEvolve have found new, faster algorithms (for example for matrix multiplication), which are new mathematical results.

Conclusion: computers can discover and prove new results in well-defined areas, usually working with human mathematicians. Creating deep, original theories across all of mathematics remains beyond them.

3. Writing an Intentionally Funny Story

Answer: Partly. Computers can write stories that are meant to be funny, but they do not reliably succeed.

Large language models can produce jokes, puns, parodies and humorous short stories on request. Early programs such as JAPE (puns) and later neural systems could generate simple jokes. However, humour depends on surprise, timing, cultural context and shared experience, so the results are often formulaic or only mildly amusing, and quality varies from one output to the next. Humans still judge them as less consistently funny than skilled comedy writers.

Conclusion: possible today, with moderate quality. Reliably producing a truly funny, original story remains an open challenge.

4. Translating Spoken English into Spoken French

Answer: Yes. This is largely solved for everyday use.

Speech-to-speech translation combines speech recognition, machine translation and speech synthesis. Real-time systems have existed since Skype Translator (2014), and today's tools such as Google Translate's conversation mode, Meta's Seamless models and live-translation features in phones and earbuds translate English speech into spoken French with little delay. Accuracy is high for common conversation but can drop with heavy accents, background noise, slang, idioms and specialist vocabulary.

Conclusion: solved well enough for practical use, though not yet equal to an expert human interpreter in all situations.

5. Performing a Complex Surgical Operation

Answer: Not yet, not fully. Computers can assist surgeons and perform parts of procedures autonomously only in laboratory settings.

Widely used robots such as the da Vinci system are controlled by human surgeons and do not operate on their own. In 2022, the STAR robot performed autonomous soft-tissue suturing in pigs. In 2025, Johns Hopkins researchers reported SRT-H, a robot that carried out the key steps of a gallbladder removal fully autonomously on eight removed pig gallbladders, with a 100% success rate across the steps, although it was slightly slower than a human surgeon. These were controlled, non-living experiments. Fully autonomous complex surgery on living human patients has not been achieved, because of safety, anatomical variation, unexpected events such as bleeding, and legal and ethical requirements.

Conclusion: not solved. Autonomous surgery works in the lab, and AI currently serves as an assistant to the human surgeon.

Overall Conclusion

Table tennis and spoken translation can be considered largely solved, theorem discovery and proof is solved in specific domains, funny story writing is partially achieved, and complex autonomous surgery remains a research goal. The tasks that are closest to being solved are those with clear rules and measurable success, while tasks needing common sense, creativity, or life-critical reliability are the hardest.

References

- Dürr, P. et al. "Outplaying elite table tennis players with an autonomous robot." Nature 652, 886–891 (2026).
- Kim, J. W. et al. "SRT-H: A hierarchical framework for autonomous surgery via language-conditioned imitation learning." Science Robotics (2025).
- McCune, W. "Solution of the Robbins problem." Journal of Automated Reasoning (1997).
- Russell, S. and Norvig, P. Artificial Intelligence: A Modern Approach (Chapter 1 exercise on tasks computers can currently solve).